SSH

This quick reference cheat sheet provides various for using SSH.

Getting Started

Connecting

Connect to a server (default port 22)

```
$ ssh root@192.168.1.5
```

Connect on a specific port

```
$ ssh root@192.168.1.5 -p 6222
```

Connect via pem file (0400 permissions)

```
$ ssh -i /path/file.pem root@192.168.1.5
```

See: SSH Permissions

Executing

Executes remote command

```
$ ssh root@192.168.1.5 'ls -l'
```

Invoke a local script

```
$ ssh root@192.168.1.5 bash < script.sh
```

Compresses and downloads from a server

```
$ ssh root@192.168.1.5 "tar cvzf - ~/source" > output.tgz
```

SCP

Copies from remote to local

```
$ scp user@server:/dir/file.ext dest/
```

Copies between two servers

```
$ scp user@server:/file user@server:/dir
```

Copies from local to remote

```
$ scp dest/file.ext user@server:/dir
```

Copies a whole folder

```
$ scp -r user@server:/dir dest/
```

Copies all files from a folder

```
$ scp user@server:/dir/* dest/
```

Copies from a server folder to the current folder

```
$ scp user@server:/dir/* .
```

Config location

`/etc/ssh/ssh_config`	System-wide config
`~/.ssh/config`	User-specific config
`~/.ssh/id_{type}`	Private key
`~/.ssh/id_{type}.pub`	Public key
`~/.ssh/known_hosts`	Logged in host
`~/.ssh/authorized_keys`	Authorized login key

SCP Options

scp -r	Recursively copy entire directories
scp -C	Compresses data
scp -v	Prints verbose info
scp -P 8080	Uses a specific Port
scp -B	Batch mode (Prevents password)
scp -p	Preserves times and modes

Config sample

```
Host server1
    HostName 192.168.1.5
    User root
    Port 22
    IdentityFile ~/.ssh/server1.key
```

Launch by alias

```
$ ssh server1
```

See: Full Config Options

ProxyJump

```
$ ssh -J proxy_host1 remote_host2
```

```
$ ssh -J user@proxy_host1 user@remote_host2
```

Multiple jumps

```
$ ssh -J user@proxy_host1:port1,user@proxy_host2:port2 user@remote_host3
```

ssh-copy-id

```
$ ssh-copy-id user@server
```

Copy to alias server

```
$ ssh-copy-id server1
```

Copy specific key

```
$ ssh-copy-id -i ~/.ssh/id_rsa.pub user@server
```

SSH keygen

ssh-keygen

```
$ ssh-keygen -t rsa -b 4096 -C "your@mail.com"
```

-t	Type of key
-b	The number of bits in the key
-C	Provides a new comment

Generate an RSA 4096 bit key with email as a comment

Generate

Generate a key interactively

```
$ ssh-keygen
```

Specify filename

```
$ ssh-keygen -f ~/.ssh/filename
```

Generate public key from private key

```
$ ssh-keygen -y -f private.key > public.pub
```

Change comment

```
$ ssh-keygen -c -f ~/.ssh/id_rsa
```

Change private key passphrase

```
$ ssh-keygen -p -f ~/.ssh/id_rsa
```

Key type

- rsa
- ed25519
- dsa
- ecdsa

known_hosts

Search from known_hosts

```
$ ssh-keygen -F <ip/hostname>
```

Remove from known_hosts

```
$ ssh-keygen -R <ip/hostname>
```

Key format

- PEM
- PKCS8

Also see

OpenSSH Config File Examples (cyberciti.biz)
ssh_config (linux.die.net)

Related Cheatsheet

Mitmproxy Cheatsheet
Quick Reference

Netcat Cheatsheet
Quick Reference

Netstat Cheatsheet
Quick Reference

Recent Cheatsheet

Remote Work Revolution Cheatsheet
Quick Reference

Homebrew Cheatsheet
Quick Reference

PyTorch Cheatsheet
Quick Reference

Taskset Cheatsheet
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